Code No: 118AA **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech IV Year II Semester Examinations, June - 2024
AD HOC AND SENSOR NETWORKS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a) Mention any two important characteristics of a Mobile ad hoc Network (Manet). [2]
b) State the main difference between topology-based and position-based Manet routing algorithms. [3]
c) What is meant by geocast in Manets? [2]
d) List the three issues of broadcast-storm problem. [3]
e) State the mathematical relationship between sensing range and communications range of a sensor node. [2]
f) What are the main components of a wireless sensor node? [3]
g) What is the objective of CONFIDANT algorithm? [2]
h) What is meant by secure routing in Manets? [3]
i) Write the important features of TinyOS operating system. [2]
j) What are the main differences between nesC and C? [3]

PART – B

(50 Marks)

2.a) Write about DSDV routing algorithm. Mention its pros and cons.
b) Differentiate AODV and DSR routing protocols. [5+5]

OR

3.a) What is meant by DREAM location service? Write about it.
b) Illustrate the mechanism of LAR (Location-Aided Routing) protocol. [5+5]

4.a) Briefly write on the rebroadcasting design considerations: Jitter and RDT.
b) Give the algorithmic steps for MPR (Multi Point Relaying) algorithm. [5+5]

OR

5.a) What are the differences between tree-based and mesh-based multicast routing protocols?
b) Write about MGR geocast protocol for Manets. [5+5]

6.a) Give the architecture used in Great Duck Island Application of sensor networks.
b) Draw the diagram for generalized architecture of wireless sensor network applications.[5+5]

OR

7.a) Write about the SMAC, a MAC protocol for WSNs.
b) What is the objective of LEACH protocol? Explain it. [5+5]

8.a) What are the characteristics a security solution for ad hoc networks should satisfy?
b) Illustrate how key agreement is achieved using Diffie-Hellman protocol. [5+5]

OR

9.a) Give the architectural diagram for IDS in ad hoc networks.
b) What is the need for enforcing cooperation in Manets? What are the approaches? [5+5]

10.a) What are the salient features of Network Simulator version 2 (NS2)?
b) Write about the methodology to be used to write code using nesC language. [5+5]

OR

11. Write short notes on the following:
a) TinyGALS
b) TOSSIM. [5+5]

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